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Review Article

An overview of Hydrogen assisted (Direct) recycling of Rare earth permanent magnets

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ABSTRACT

Rare Earths (RE) permanent magnets are essential components for Europe's successful green and digital transition However, the entire value chain of RE magnetic materials depends on imports, which are highly vulnerable in current global supply chain models. To mitigate this situation, EU Regulation plans that at least 15 % of the EU's annual consumption of permanent magnets should be covered by recycling capacities by 2030. Researchers in the EU H2020 project SUSMAGPRO consortium have shown that hydrogen can be used as a very efficient recycling method to extract NdFeB magnet powder from various EOL Components in the IP protected Hydrogen-based Processing of Magnet Scrap (HPMS). On exposure to hydrogen the sintered NdFeB magnets break down into a friable, demagnetised, hydrogenated powder containing an interstitial hydride of $Nd_2Fe_{14}BH_X$ (10 μm) and smaller particles (<1 μm) from the grain-boundary phase $NdH_{2.7}$. This process delivers a sustainable source of magnetic material for the production of sintered, polymer bonded and metal-injection moulded magnets. The paper will present numerous results along the whole value chain of magnet recycling, including automatic dismantling of magnet containing products, magnets extraction, HPMS recycling, production of recycled magnets and demonstrator testing. It will also discuss best practices and bottlenecks of the processes as an outlook for successful design-for-recycling of future applications.

1. Introduction

Making the EU a climate-neutral, circular and competitive economy by 2050 depends on Europe's own ability to develop and deploy clean energy and mobility solutions in an economically and environmentally sustainable way [1]. Industrial and household appliances will also need to operate to the highest energy efficiency standards. Electric motors and generators, which convert electrical energy into motion (or vice versa), are essential components of these applications, particularly in electronics, communications and medical equipment, renewable energy, robotics, electric vehicles, aerospace and dual use. The most efficient motors and generators contain permanent magnets (PMs) made of RE, i.e. neodymium, praseodymium, dysprosium and terbium, which have been identified as critical raw materials [2].

While the use of RE for magnets accounts for only 25 % of total RE production volume, it represents 80–90 % of the total market value of RE [3]. The market for RE magnets itself is relatively small - around €6.5 billion - but its downstream leverage is huge: the mobility business in the EU27 alone is expected to grow to around €500 billion by 2030, supporting six million jobs [4]. The EU's Joint Research Centre has modelled climate targets for 2030 and 2050 and translated them into annual raw material demand. The analysis shows that for the three key sectors of renewable energy, electromobility and defence and space, we will need up to twice as much neodymium/praseodymium and five times the amount of dysprosium in 2030 and almost four times as much neodymium/praseodymium and 13 times more dysprosium in 2050 compared to the current supply for the whole EU economy.

While the EU is a major player in the production of magnet

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containing components for aforementioned applications, it is completely dependent on imports along the entire value chain of RE magnet materials. Of the approximately 140,000 tonnes of RE permanent magnets (REPM) produced worldwide in 2021, 93 % were manufactured in China, reflecting a very high concentration of production (which is indeed present at all stages of the RE value chain, from mining to recycling). Today, there is only 1,000 t of magnet production capacity left in Europe, competing with 16,000 t of imported Chinese magnets per year. Despite a growing market, European magnet production capacity is under-utilised and tends to serve specialised niche applications. In addition, RE magnets are increasingly being imported as part of motor and generator assemblies and products. The main reasons for these developments are that China has a monopoly in the RE supply chain at all stages from mining to refining, and that Chinese material prices do not reflect the real cost of production as they are kept artificially low by direct and indirect public subsidies.

In its flagship report [5], the European Rare Earths Competence Network highlighted the dangers associated with European industry's total dependence on imported materials. Without considering trade wars or import tariffs, the COVID-19 pandemic has highlighted further vulnerabilities in current global supply chain models, particularly where key European industries rely on a small number of third country suppliers for materials and components. As a result, there is a lack of supply chain diversification, market-driven competition and resilience to supply shocks; supply chain transparency and clear definition of sustainability standards; industrial capacity and uptake of innovation to implement a European circular economy for RE used in magnets; and strategic investment to benefit from a growing materials market to create economic growth and new jobs in Europe directly related to the value chain of RE magnets and motors, as well as indirect jobs secured downstream, for example by maintaining strong industrial leadership in electric vehicle manufacturing in Europe. The need for a diversified material supply and improved manufacturing capabilities is clearly described in the recent European Commission study on critical raw materials [2]. Achieving the climate and digital transitions and developing the EU's open strategic autonomy depends on the availability of sustainable and responsibly sourced critical raw materials.

2. Recycling

2.1. Strategies

Depending on the stage at which the recycled or EoL magnetic products enter the material flow, recycling can be based on one of the following strategies:

(1) Material recycling, where scrap magnets are fed into smelting processes as raw materials;
(2) Element recycling, where the RE are extracted from the scrap magnets and regenerated in high purity for re-alloying; and.
(3) Magnet recycling, in which the scrap magnets are treated to maintain the chemical composition of the alloy and the resulting material can be used directly for the remanufacture of recycled magnets..

Hydrometallurgical routes [6–11] require large quantities of chemicals and water and involve energy-intensive processing steps with long total processing times. Large amounts of waste water are generated and not only RE metals but also Fe is dissolved by acid solutions. Fe and B residues have to be considered as industrial waste because the recycling costs of Fe do not correspond to the market price and B is a toxic element that needs to be controlled environmentally [12].

Pyrometallurgical (high temperature) routes [13–15] have been developed as an alternative to hydrometallurgical routes and have lower water consumption and waste generation, but also high energy consumption. Some pyrometallurgical routes allow the remelting of REE alloys or the extraction of REEs from transition metals in the metallic state; other routes can be used to recycle partially oxidized REE magnet alloys.

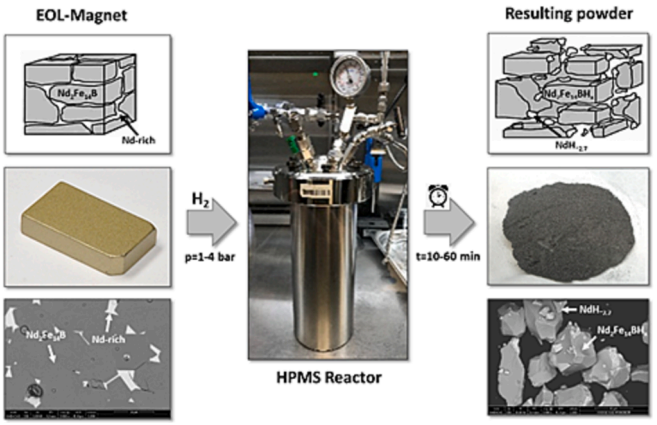


Fig. 1. HPMS process (direct- or "short-loop" recycling) for NdFeB-type magnets.

The advantages are that RE metal can be obtained by separating the REEs from the waste, and highly concentrated impurities can be removed. Disadvantages include long processing times, high energy requirements and high environmental impact [12].

Physical/mechanical separation technologies are mainly used to separate REPM from other wastes in strategy (3) and also to support strategies (1) and (2) [12]. The respective technologies are elaborated further in sections 3.2 and 3.3.

2.2. Hydrogen processing of magnetic scrap

The direct-recycling of magnets (or "short-loop recycling") (3) into sintered magnets, which involves less energy-intensive processing steps than pyrometallurgy and hydrometallurgy [16,17], is attracting considerable interest, with promising results depending on the input of scrap NdFeB magnets [18–21]. The process is known as magnet-to-magnet recycling, "short-loop recycling" or "Hydrogen Processing of Magnetic Scrap (HPMS)"(see Fig. 1) and is a patented process developed by researchers at the University of Birmingham [22,23].

When exposed to hydrogen, the sintered Nd-FeB magnets break down into a friable, demagnetised, hydrogenated powder containing an interstitial hydride of $Nd_2Fe_{14}BH_X$ (10 μm) and smaller particles (<1 μm) of the grain boundary phase ($NdH_{\sim 2.7}$) both of which can be mechanically separated from residual coatings, adhesives, screws and fasteners.

The H2020 project SUSMAGPRO [24] demonstrates the HPMS process on a pilot scale (100 tonnes per year) by separating and purifying Nd-Fe-B powders from a wide range of robotically sorted parts and the production of materials with four different processing routes.

3. Separation of magnets from EOL applications

3.1. Material streams

Efficient recycling needs to be based on material flow analyses (MFA) that map the relevant material flows in society. For neodymium flows in the EU, four recent studies have contributed to an understanding at an aggregated level [25–29]. As different products contain Nd, the ease of recycling can vary significantly between products [28,29]. However, the information on total volumes of neodymium flows in previous MFAs is insufficient to identify recycling opportunities for three reasons: (i) inconsistencies in market shares and scope definitions create uncertainties; (ii) in what is currently a very country-specific and localised recycling market, only a European perspective on the waste flows is

Table 1
Recyclability of selected waste flows. Aspects scores are normalized to the range 0–5, where 5 is most favorable for recycling [31].

Life cycle stage		EV motors	HDDs	Industrial pumps	Speakers	Wind turbines
Overarching	Supply chain alignment	1.3	1.0	0.5	0.5	3.0
	Uncertainty	3.3	4.0	1.7	1.6	4.5
	Economic drivers	1.4	1.9	3.0	3.0	2.4
	Social benefits	4.4	4.4	4.4	4.4	4.4
Manufacturing	Design for disassembly	1.6	2.8	2.0	2.3	4.2
Use & collection	Collectability	2.2	1.6	3.5	2.2	3.6
	Policy	3.0	2.3	0.5	2.3	3.8
	Collection rate	3.1	3.5	0.8	3.0	4.5
Preprocessing	Preprocessing performance	4.3	4.0	2.9	3.2	5.0
	Safety risk	3.5	3.0	5.0	4.5	4.5
Recovery	Environmental effects	4.8	4.8	4.8	4.8	4.8
	Recovery performance	4.9	3.9	4.4	4.9	4.4
	Technology availability	2.3	2.3	2.3	2.3	2.3
Secondary market	Demand	1.9	1.9	1.9	1.9	1.9
	Revenues	5.0	5.0	4.2	3.6	3.9

available [30]; and (iii) the focus on total volumes ignores many important factors that determine the viability of recycling, such as product design and technology availability [31,32].

These issues have been addressed by the SUSMAGPRO project with a comprehensive MFA study [31] assessing the recyclability of key EoL products. It was found that different product groups determine future waste volumes: HDDs represent a large current waste stream, while the demand for magnets in industrial applications is increasing. In the future, electric vehicle motors and wind turbines are likely to provide a source of neodymium with good recyclability. To manage the changing waste streams, SUSMAGPRO is investigating neodymium recycling systems that take into account the product characteristics of future waste streams (see also Table 1). In addition, the recyclability of products will be improved by removing bottlenecks in the recycling chain.

3.2. Sorting and separation strategies

The recyclability assessment [31] showed that the recovery process, although immature, is not the main bottleneck for recycling. Rather, the barriers to commercialisation lie in design, waste collection and other product-specific steps. The main issues in recovery are (i) content heterogeneity in End-of-Life (EoL) products [33,34], which can be SmCo, Ferrite and NdFeB materials. All require different recycling strategies and a material mix is counterproductive to their recycling. In the case of NdFeB, there is even a fundamental difference in recycling technology between sintered and polymer bonded magnets, with the latter currently unsuitable for large-scale recycling. The situation is exacerbated by (ii) underdeveloped recycling schemes, with many countries having mixed recycling of WEEE and unclear regulations regarding the recycling of products containing magnets. In addition, (iii) the magnet content of products is often low, making dismantling solely for the recovery of the magnet unattractive, as the dismantling cost can be many times greater than the material value recovered. This is further accentuated by (iv) haphazard design for recycling, as specific recycling requirements are not (yet) known to the designer. In the (v) absence of labelling or product passport information on magnet recycling, (vi) most products containing magnets are shredded and current shredding processes are not suitable for recycling NdFeB permanent magnets.

As a result, magnets are currently considered more of a nuisance in the processes (jamming the shredder, agglomerating and causing downtime) than a valuable resource. In some cases, manual dismantling is carried out, but this requires labour, expertise and specialised tools and is far from being commercially viable.

3.3. Automatic sorting

Automated disassembly is therefore becoming an increasingly desirable option, with tailor-made units focusing on different value

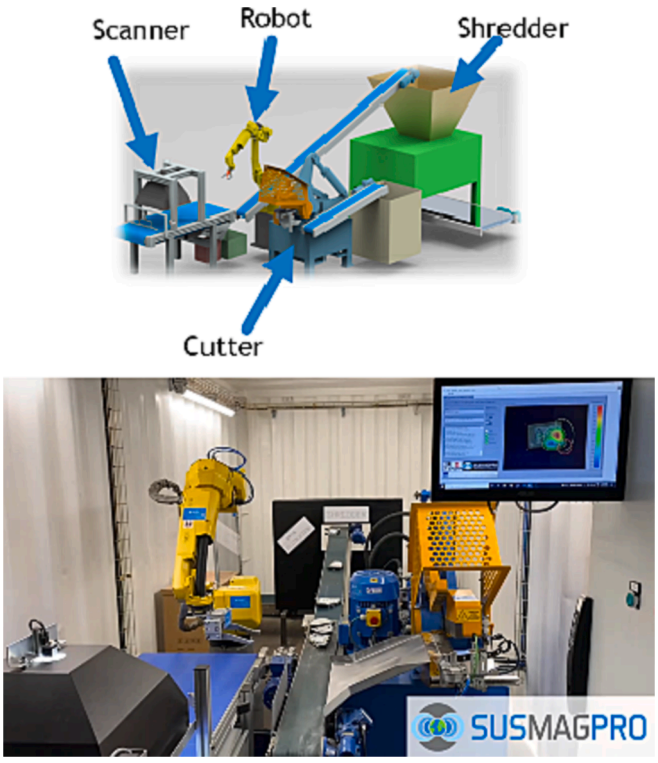


Fig. 2. SUSMAGPRO HDD pilot (schematic, left) and built inside a 20ft container (right) [40].

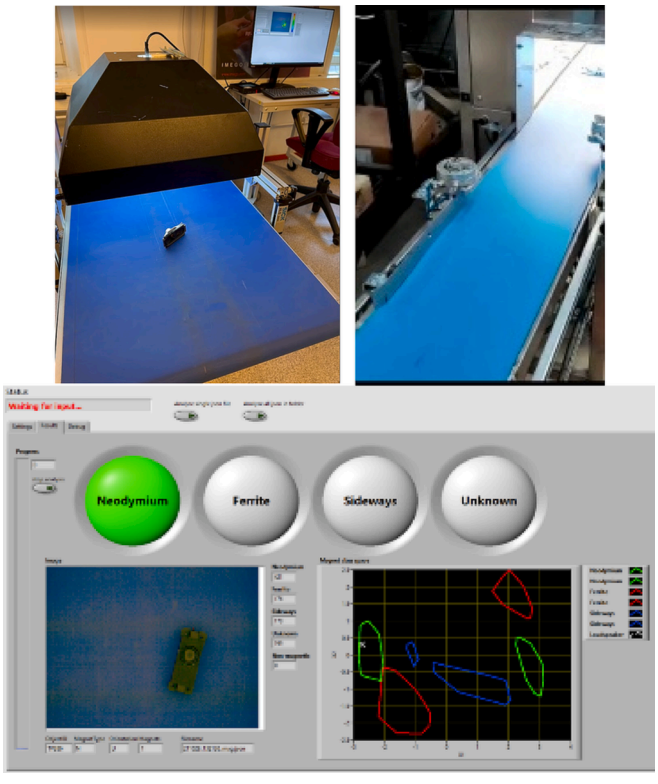


Fig. 3. SUSMAGPRO flatscreen TV pilot (left) and software frontend (right).

streams such as HDDs [35–38], industrial pumps and traction motors [39,40].

In the framework of SUSMAGPRO, four different disassembly/

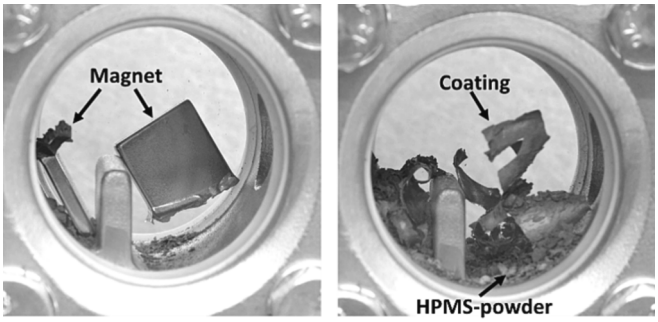


Fig. 4. HPMS process of a Ni-coated NdFeB magnet (broken into two halves) in a glass laboratory reactor (room temperature, 1 bar hydrogen pressure). Left: start of reaction, right: end of reaction, with large coating debris that can be easily separated by sieving.

sorting units have been designed and built on a pilot scale: (i) a fully automated unit for HDDs (Fig. 2), which removes the voice coil magnet prior to data destruction by shredding [41], and whose mobile design allows a significant reduction in magnet extraction time (6 s per HDD) compared to previously reported semi-automated (2 min) [38] while ensuring full compliance with the requirements of the European Data Protection Regulation (GDPR), (ii) an automated sorting unit to separate ferrite magnets from NdFeB magnets in flat screen televisions using state-of-the-art facial recognition software (Fig. 3), (iii) a pilot for automated extraction of embedded magnets from rotors of domestic heating pumps, and (iv) a pilot for semi-automated disassembly to extract and demagnetise magnets from high-end loudspeaker sets of professional P. A. systems.

All SUSMAGPRO pilots have been designed to address the low-cost, high-speed extraction of NdFeB type magnets from emerging EOL sources, including the assessment of the associated CAPEX and OPEX costs [41], thus providing valuable information for the generation of circular economy business cases. Transferability to other applications, such as traction and wind turbine motors, is also part of the investigation. The work will continue after the SUSMAGPRO project ends in November 2023 in the framework of the HORIZON project REEsilience [42].

4. Powder production

HPMS (Fig. 4.) is an environmentally friendly recycling process as it eliminates all the processing steps of conventional recycling that require large amounts of chemicals, water and energy [28,43]. However, there are certain requirements for successful HPMS recycling to produce high quality material for remanufacturing high-performance magnets. (i) The anti-corrosion coating of the EOL magnet must allow the penetration of hydrogen molecules to initiate hydration of the grain boundary phase [22], to avoid a deterioration of the magnetic properties, which can be dramatic for some residues such as nickel, even at low levels, the coating must either (iia) be mechanically separable from the generated powder (e.g. by sieving or sifting) [23], (iib) be removed prior to HPMS treatment [44], or (iic) be incorporated into the powder without affecting the magnetic properties [45].

(iii) The particle size of the HPMS powder (including agglomerates) must be reduced to ~ 3–5 μm by a subsequent milling step (preferably high energy milling/jet milling). Sintering of unground HPMS material is possible, but will result in increased grain size compared to jet-milled material and thus reduced magnetic properties. (iv) Oxygen levels (resulting from service life or processing during recycling) and carbon levels resulting from organic residues (e.g. adhesives, polymer fixings, etc.) must be kept below a certain threshold to avoid the formation of unwanted oxides and carbides [46]. While the formation of carbides reduces the coercivity, high oxygen levels lead to low density and catastrophic magnetic values as the presence of stable Nd oxides results

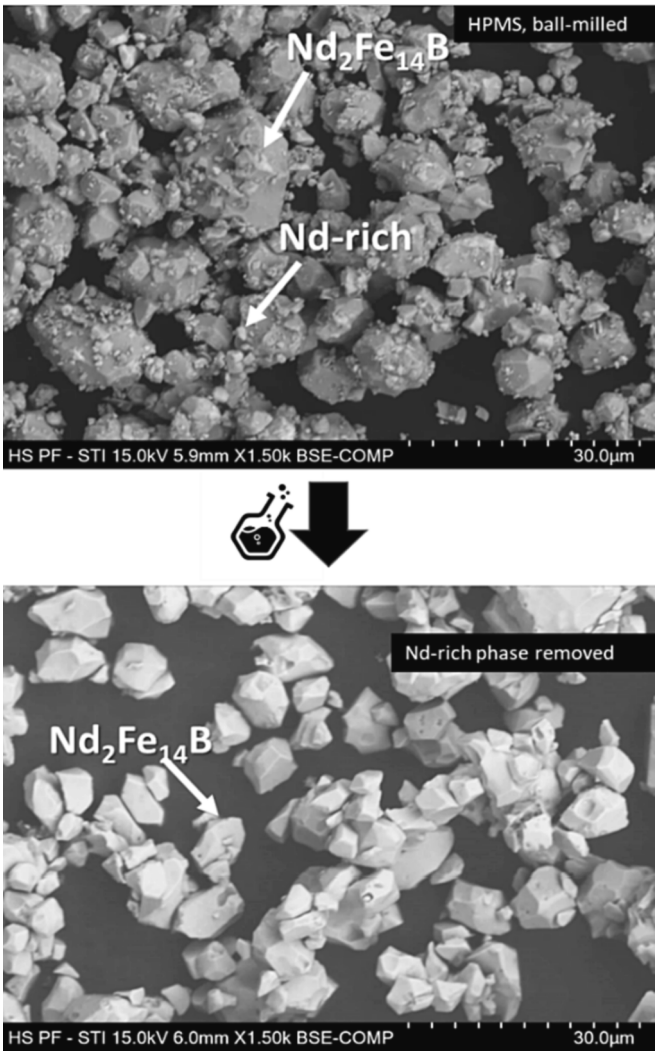


Fig. 5. Above HPMS powder, ball milled, with finely dispersed debris of the Nd-rich grain boundary phase. Below: HPMS powder after NeoLeach procedure, consisting solely of hard-magnetic ($Nd_2Fe_{14}B$) phase.

in the absence of the liquid phase required for sintering [22] (sintered magnets) or for the formation of disproportionate structures in the case of HDDR material for bonded magnets [47]. However, previous work [21,22,48] has shown that this effect can be overcome by adding additional Nd in the form of NdH_x.

As the particle size of GBP is too small to be separated by sieving or conventional screening technology, a mechanical approach to remove impurities and oxides from GBP is to employ hydrocyclone treatment [49,50], preferably integrated into the jet milling process for processing and cost reasons.

Very recent work addresses the issue of C and O impurities and chemical variations in the HPMS material by a consistent leaching step ("NeoLeach" that completely removes the SE-rich grain boundary phase (GBP) of the material [51,52].

This approach aims at the complete recovery of the contaminated GBP by electro-refining and blending the remaining HPMS powder, which now consists only of the hard magnetic phase ($Nd_2Fe_{14}B$) with new grain boundary material (e.g. virgin NdH_x or Nd70Cu30 low-eutectic alloy acting as a liquid-phase sintering aid and grain-boundary modifier), allowing high remanence values and complete reengineering of the GBP [53].

Work in SUSMAGPRO has also shown that HPMS material of different EoL sources and respectively of different chemical

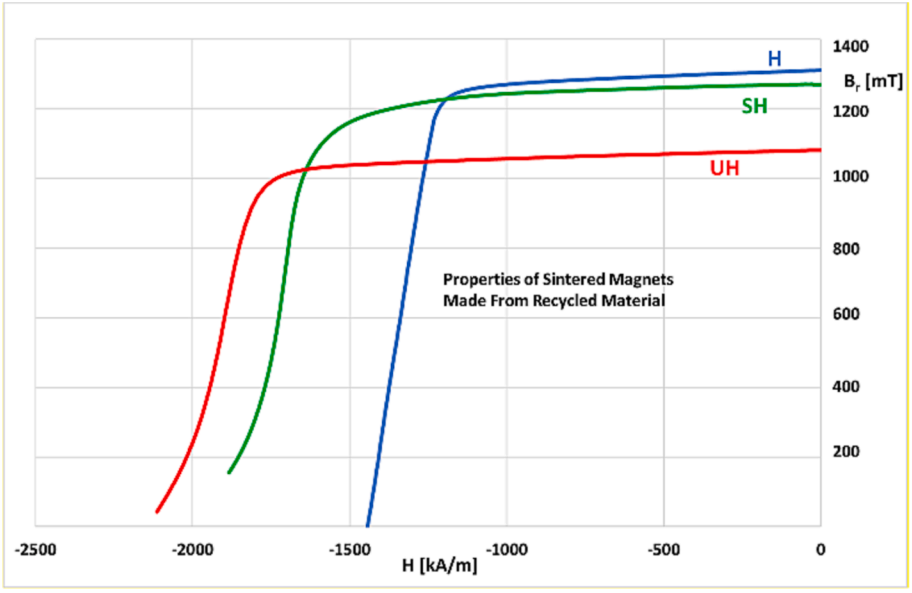


Fig. 6. Properties of recycled magnets sintered in the framework of SUSMAGPRO from HPMS material.

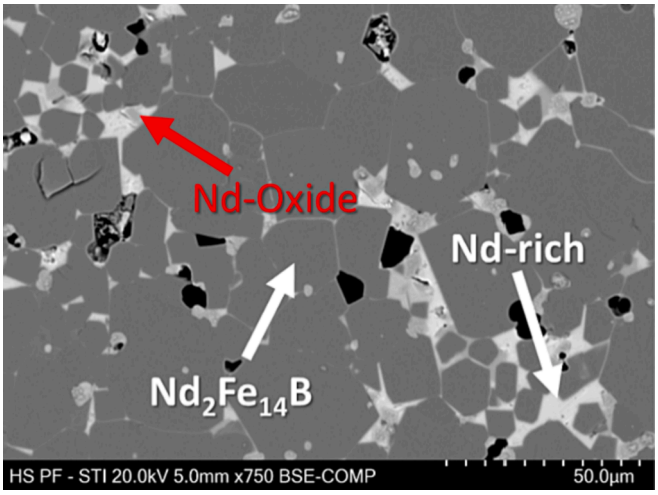


Fig. 7. Microstructure of.a recycled NdFeB magnet from HPMS material, showing hard magnetic phase ($Nd_2Fe_{14}B$), and Nd-oxides, which are embedded in the Nd-rich areas.

compositions can be blended to fine-tune the magnetic grades [54].

5. Magnet production using HPMS-recyled powders

5.1. Sintering

The sintering process for HPMS powder (secondary material) is identical to that for primary material from powder (jet) milling onwards. Furthermore, any ratio between primary and secondary material is possible, which can be used to fine-tune the composition and can be very beneficial for a high recycling rate of contaminated HPMS material, which can be "diluted" with primary material, a strategy further developed in the Horizon Europe REEsilience project. In SUSMAGPRO, a variety of magnetic grades have been manufactured, see Fig. 6.

NdH_x (1–5 wt%) and Dy (up to 4 wt%) have been added to fine-tune the properties. It can be seen that very high coercivity values can be achieved. However, the maximum remanence values are all below 1.3 T due to the presence of varying amounts of NdO in the material (see Fig. 7).

It can be seen that the Nd oxides are embedded in the Nd-rich regions, indicating that they are enveloped by the Nd liquid phase during the sintering process.

It should be noted that the results of SUSMAGPRO recycled magnets shown in Fig. 6 correspond to commercially available high-performance grades which are already suitable for a wide range of applications.

The use of strategies such as the NeoLeach process combined with appropriate grain boundary engineering, where heavy rare earth such as Dy and Tb are distributed in thin layers around the GBP [55–58] will allow the current limitations to be overcome and the production of N48 + grade magnets is expected in the near future.

5.2. Polymer bonding

The Hydrogen Disproportionation Desorption and Recombination (HDDR) process route has been shown to be an effective method of reprocessing scrap sintered magnets [19,22,59]. In SUSMAG-PRO, polymer bonded magnets were produced from HPMS material under near-serial conditions. The bonding material was PA12 plus ~ 1 % additives (compatibilizers, lubricants) with a filling ratio 58 vol% magnetic material (HDDR-ed NdFeB), injected at 270 °C with a pressure of 880–930 bar in an alignment field of 750 mT..

As can be seen from Fig. 8, magnets with Br of 580 mT and iHc of ~ 680 kA/m could be produced. The relatively low remanence indicates that higher orienting fields are required to manufacture fully anisotropic magnets, an improvement in coercivity and squareness is targeted with optimised HDDR processing parameters. In parallel, the production of melt-spun polymer-bonded magnets using near-stochiometric NeoLeach material (see Fig. 5) is an alternative development route being pursued for the production of recycled magnets from HPMS powder.

5.3. Metal injection moulding

To produce anisotropic MIM magnets from recycled material, HPMS powder is mixed with a multi-component binder and then injection moulded into a cavity with an integrated magnetic field to form a so-called "green part". A combination of solvent and thermal debinding is used to remove the binder, followed by a sintering step in an inert gas or vacuum atmosphere to form the final, now binder-free magnet [60,61]. In SUSMAGPRO, MIM magnets with B_r up to 1.25 T and ${}_iH_c$ of 1450 kA/m have been produced with a slightly higher C content

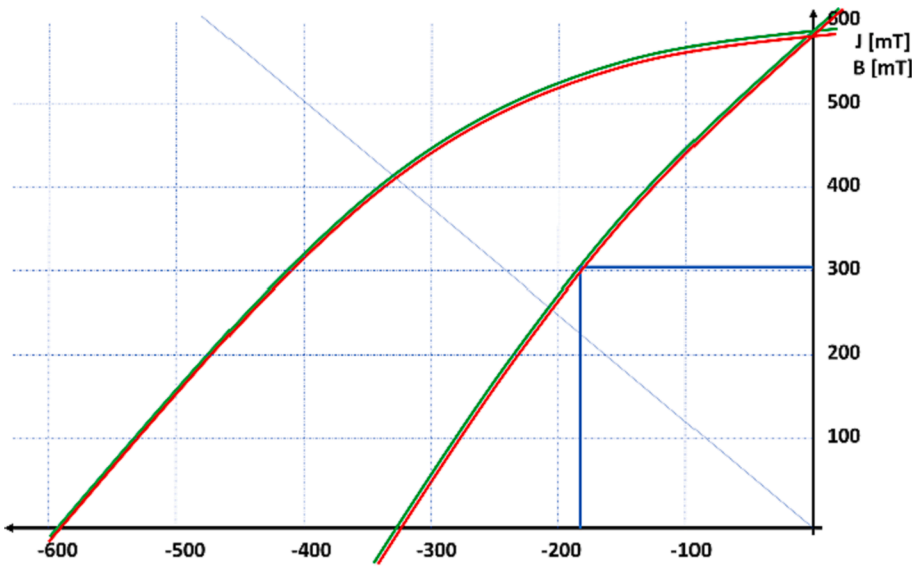


Fig. 8. Properties of recycled polymer bonded magnets produced in the framework of SUSMAGPRO from HPMS material followed by HDDR treatment (work in progress).

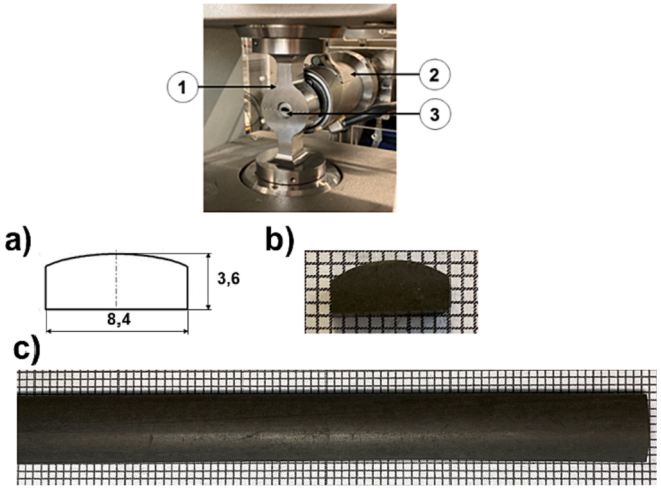


Fig. 9. above: PEM–mould with magnetic alignment field (1) magnetic alignment (2) heating element (3) nozzle; below: geometry, cross-section and strand of PEM-magnet.

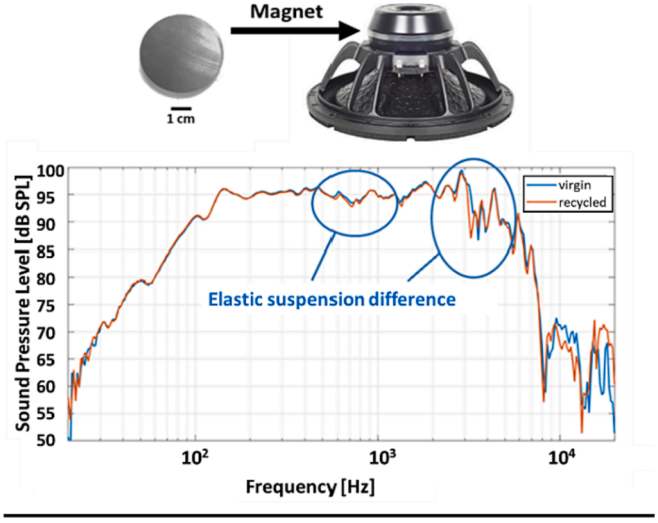


Fig. 10. Above: High-end loudspeaker system with HPMS recycled sintered magnet. Below: Comparison of frequency response of recycled vs. original magnet.

compared to sintered magnets made from identical HPMS material. This is due to residues of the organic backbone binder that could not be completely removed during the debinding process.

Although MIM magnets made from recycled material currently have slightly inferior properties compared to sintered magnets, their use in small and miniature components could be promising: when a sintered magnet is shaped, the heat input from cutting and grinding processes degrades the magnetic properties on the machined surfaces, which is disproportionately important for miniature structures where the surface-to-body ratio is particularly high [53].

Since miniature MIM magnets do not require such post-treatment, the initial poorer magnetic properties may be compensated for, depending on the final geometry.

In SUSMAGPRO, a novel approach to produce anisotropic NdFeB magnets via powder extrusion moulding (PEM) in a continuous process [60] is investigated, see Fig. 9. Samples with ${}_iH_C$ of up to 1.500 kA/m have been produced, however so far, no sufficient magnetic alignment could be achieved so far, producing isotropic magnets with a B_r of only 550 mT. The development of a continuous alignment system for production of anistropic PEM-magnets is subject to further investigation.

6. Demonstrators

As part of the SUSMAGPRO demonstration activities, high-end loudspeaker systems were assembled using short-loop recycled sintered magnets, as described in sections 4 and 5.1, manufactured to the OEM's specifications (geometric and magnetic values). All samples were assembled under production conditions to eliminate the possible effects of prototype procedures. The audio systems were tested for performance against systems using original magnets on state-of-the-art equipment, testing.

(i) frequency response, i.e. the sound pressure produced by the loudspeaker when supplied with a sinusoidal stimulus of constant amplitude at all frequencies, (ii) nonlinear characteristics, in particular the dependence of the B field in the gap with position, to detect potential

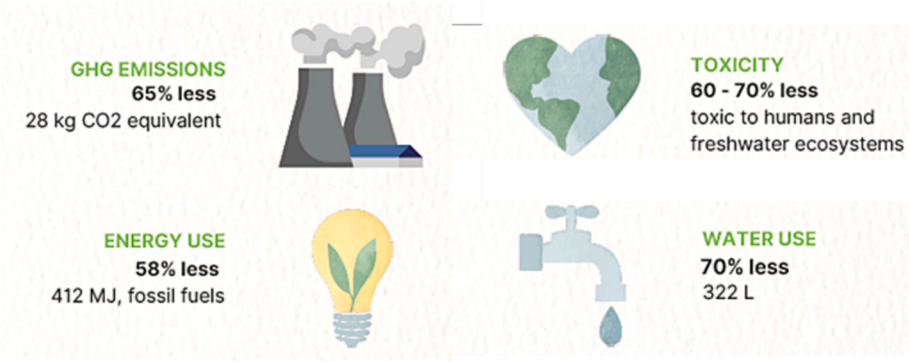


Fig. 11. Environmental impact of 1 kg sintered loudspeaker magnet (HPMS short loop) compared to 1 kg of equivalent primary magnet [50].

differences in the magnetic properties and (iii) long-term behaviour of the loudspeaker, with particular attention to a potential thermal degradation.

No difference in performance could be detected between the systems using recycled and original magnets (see also Fig. 10), providing a proof of concept for the HPMS "short loop" recycling method at demonstrator level. It should be noted, that the measurement differences in frequency response are attributed to the elastic suspension difference in the membrane system and are within normal serial deviation [62].

As part of a comprehensive Life Cycle Assessment (LCA)[1] carried out within the framework of SUSMAGPRO, the key environmental values for the production of sintered magnets for the above-mentioned loudspeaker production have been compared for both the "short loop" HPMS route and the conventional primary route, see Fig. 11.

It should be noted that the current values for the HPMS route reflect the use of prototype equipment for dismantling EOL loudspeaker systems and powder production, which is not yet optimised in terms of energy, cooling water and protective gas consumption [54]. This suggests that even better environmental performance is possible under production conditions.

The SUSMAGPRO project is currently testing other demonstrators of traction motors, heat pumps and sensor systems using a similar approach.

7. Design for recycling

As described above, much progress has been made in developing technologies to identify, separate and reprocess magnets from waste streams. However, a major remaining obstacle is the sourcing and dismantling of REPM-containing scrap. The dismantling process needs to be optimised, particularly in terms of maximising the yield of high-quality material while being economical. This is where efficient eco-design (design for recycling) can make a massive difference.

A comprehensive guide for traction motor applications has been developed in SUSMAGPRO [53] and approaches for other applications are underway. In general, knowledge of current coatings and fixation methods will greatly improve the chances of efficient recycling, while recognising that new recycling technologies may be developed between the design of current magnet-containing products and their end-of-life. Adequate, standardised labelling will be essential to maximise effectiveness [1,46].

[1] The LCA accounts for all processes starting from the liberation of the magnets from EOL product, including transport to the HPMS powder/magnets manufacturer. Collection and transportation of waste are not taken into account, as it is assumed that the waste products are already collected independent of magnets recycling.

8. Conclusion

Hydrogen Processing of Magnetic Scrap (HPMS) as an environmentally friendly, energy efficient alter-native to hydrometallurgical and pyrometallurgical recycling of NdFeB permanent magnets has made tremendous progress on both the development and manufacturing side and is now on the verge of commercialisation. While the core technology (production of powder from scrap magnets) is mature and ready for use, there are still parts of the value chain that need to be further improved for a true, cost-effective circular economy approach. This applies in particular to scrap collection and dismantling, as well as efficient eco-design (design for recycling). There is a need to raise awareness among users and manufacturers of components containing magnets, and this must be accompanied by legal requirements for take-back, recycling rates, labelling obligations, traceability and recycling quotas in order to achieve a true circular economy for permanent magnets.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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